National University of Computer and Emerging Sciences
Lahore Campus

Engineering Economics (MG3008)

Sessional-II Exam

Date: April 6 2026

Course Instructor(s)

Mr. Syed Haris Mujtaba Kazmi

Total Time (Hrs):	**1**
Total Marks:	**75**
Total Questions:	**3**

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Roll No Section

[redacted]

Student Signature

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CLO # 2: Interpret the concept of time value of money

Q1: Machines that have the following cost are under consideration for a robotized welding process. Using an interest rate of 10% per year, **Determine** which alternative should be selected on the basis of future worth analysis. **[Marks: 25]**

	Machine X	Machine Y
First cost, $	-250,000	-430,000
Annual operating cost, $/year	-60,000	-40,000
Salvage value, $	70,000	95,000
Life, years	3	7

CLO # 2: Interpret the concept of time value of money

Q2: A total of $100,000 was invested in two different projects identified as Z1 and Z2. If the overall rate of return on the $100,000 was 30% and the rate of return on the $30,000 invested in Z1 was 15%. **Calculate** the rate of return on Z2. **[Marks 25]**

CLO # 3: Analyze cost effectiveness of alternatives using engineering economy methods

Q3: Four methods are available to recover the lubricant from an automated milling system. The investment cost and income associated with each other are shown. Assuming that all methods have a 10-year life with zero salvage value, **Compare** the given methods and determine which one should be selected using a MARR of 10% per year and the incremental benefit cost method. Consider annual operating cost as a disbenefit. **[Marks 25]**

	1	2	3	4
First cost, $	15,000	19,000	25,000	33,000
Annual operating cost, $/year	10,000	12,000	9,000	11,000
Annual income, $/year	16,000	20,000	19,000	22,000

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Formula sheet

$\left(\frac{F}{P}, i, n\right) => F = P(1+i)^n$	$\left(\frac{P}{F}, i, n\right) => P = \frac{F}{(1+i)^n}$
$\left(\frac{P}{A}, i, n\right) => P = A\frac{(1+i)^n - 1}{i(1+i)^n}$	$\left(\frac{A}{P}, i, n\right) => A = P\frac{i(1+i)^n}{(1+i)^n - 1}$
$\left(\frac{F}{A}, i, n\right) => F = A\frac{(1+i)^n - 1}{i}$	$\left(\frac{A}{F}, i, n\right) => A = F\frac{i}{(1+i)^n - 1}$
Arithmetic Gradient	$P = \frac{G}{i}\left[\frac{(1+i)^n - 1}{i(1+i)^n} - \frac{n}{(1+i)^n}\right]$
Arithmetic Gradient	$A = G\left[\frac{1}{i} - \frac{n}{(1+i)^n - 1}\right]$
Geometric Gradient $P_g = A_1\left\{\frac{1-\left(\frac{1+g}{1+i}\right)^n}{i-g}\right\} \quad g \neq i$	$P_g = A_1\left(\frac{n}{1+i}\right) \quad g = i$
$Effective i = (1 + {}^{r}/_{m})^m - 1$	
Inflation Adjusted Interest Rate	$i_f = i + f + if$
$AW_k = -P(A/P,i,k) + S_k(A/F,i,k) - \left[\sum_{j=1}^{j=k} AOC_j(P/F,i,j)\right](A/P,i,k)$	